Foliar fungal symbionts in sympatric yellow monkeyflowers along elevation gradients in the Sierra Nevada

Bolívar Aponte Rolón[1,2,✉], Kathleen G. Ferris[2], and Sunshine A. Van Bael[2]

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[1] Department of Agricultural and Biosystems Engineering, Iowa State University, Ames, IA 50011, USA

[2] Department of Ecology and Evolutionary Biology, Tulane University, 6823 St. Charles Avenue, New Orleans, LA 70118

✉ Correspondence: Bolívar Aponte Rolón <bolivar@iastate.edu>

13. Supplementary Material

13.1 Figure S1

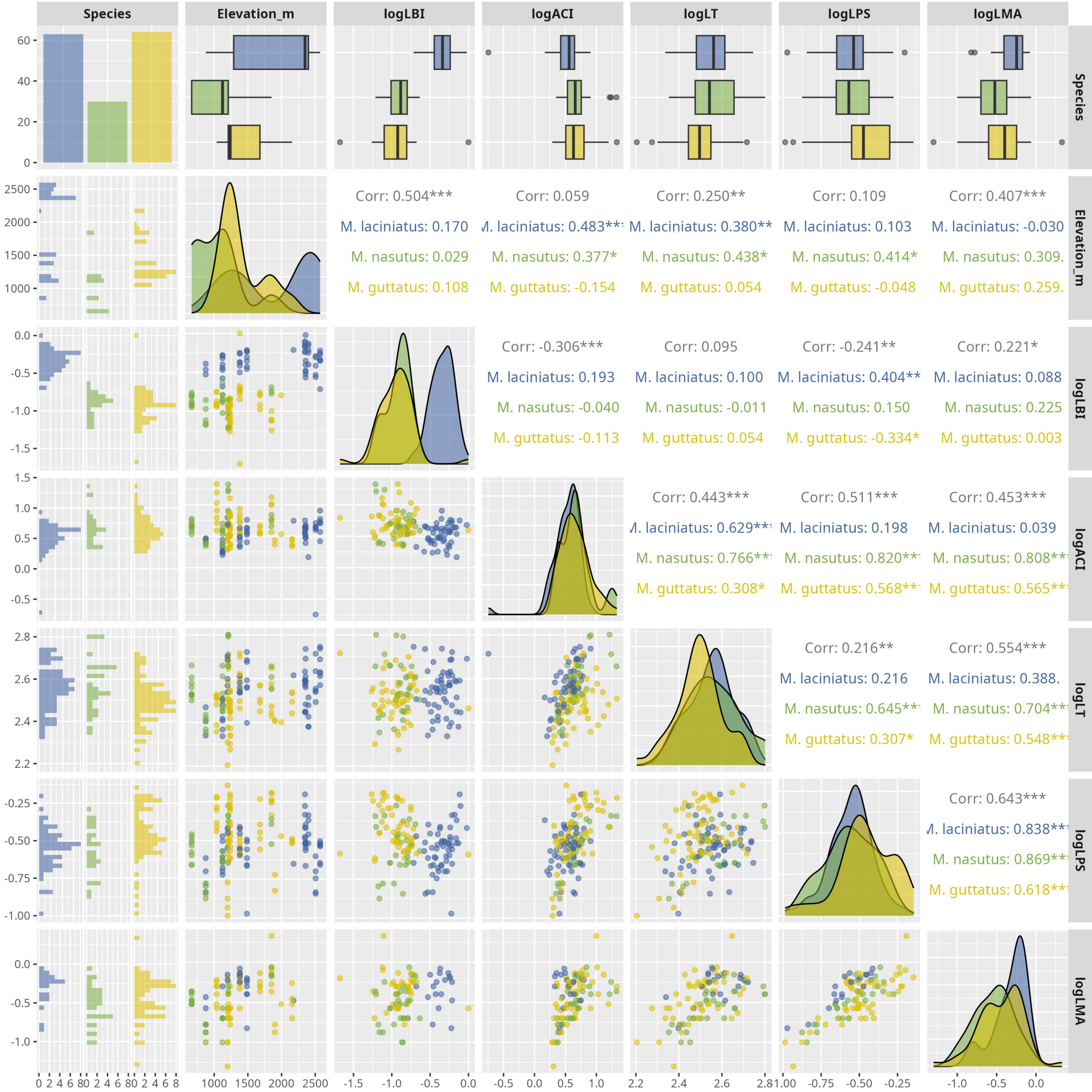

Figure S1. Correlation matrix of log-transformed leaf functional traits by host species. The plot is based on the Spearman's rho. Significance levels are represented by *ns* (not significant) and asterisks [$p < .05$ (*), $p < 0.01$ (**), $p < .001$ (***), and $p < .0001$ (****)].

13.2 Figure S2

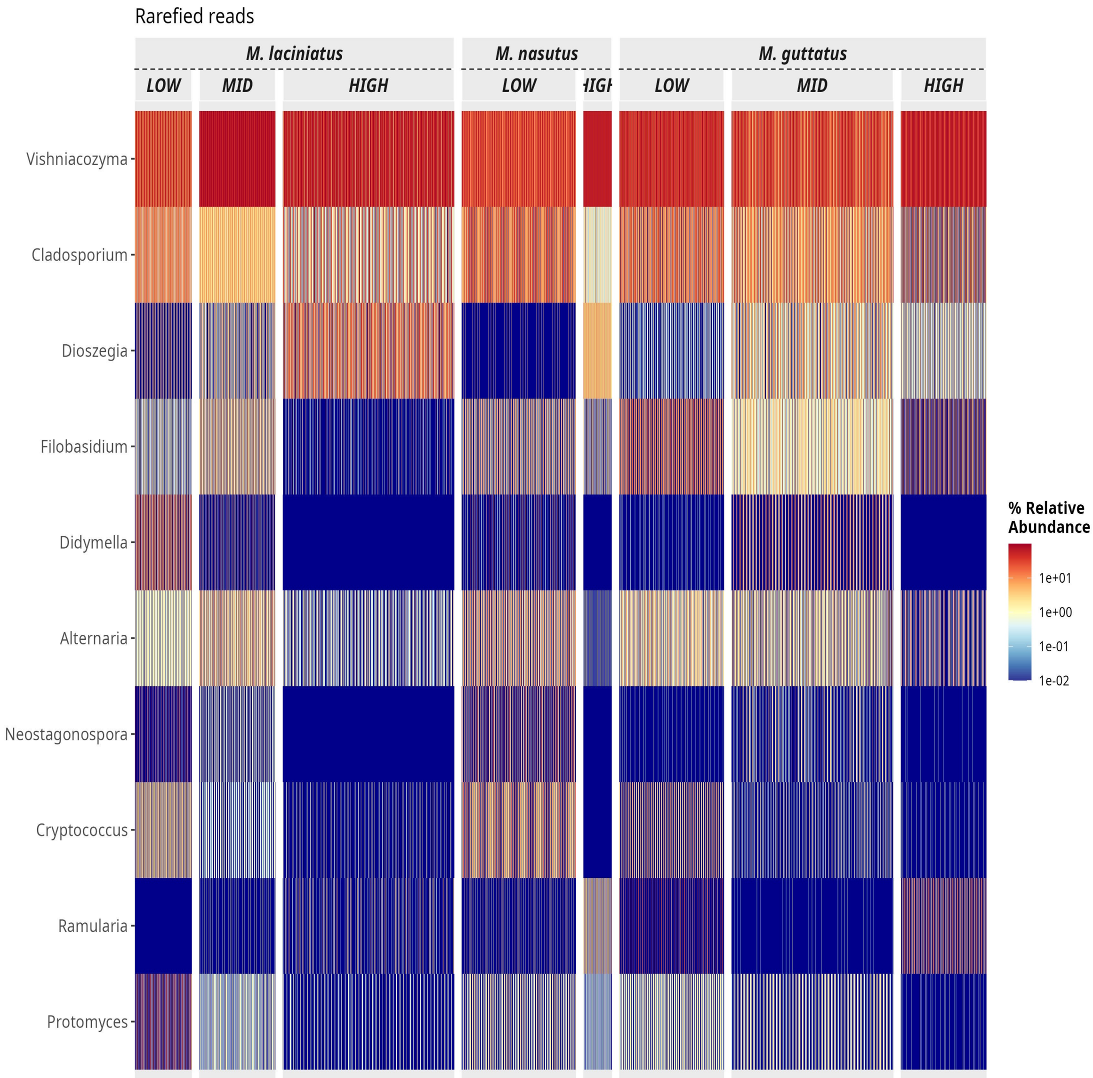

Figure S2. Heatmap of the top 10 most abundant FEF genera in the dataset. The heatmap is faceted by host species and elevation zone.

13.3 Figure S3

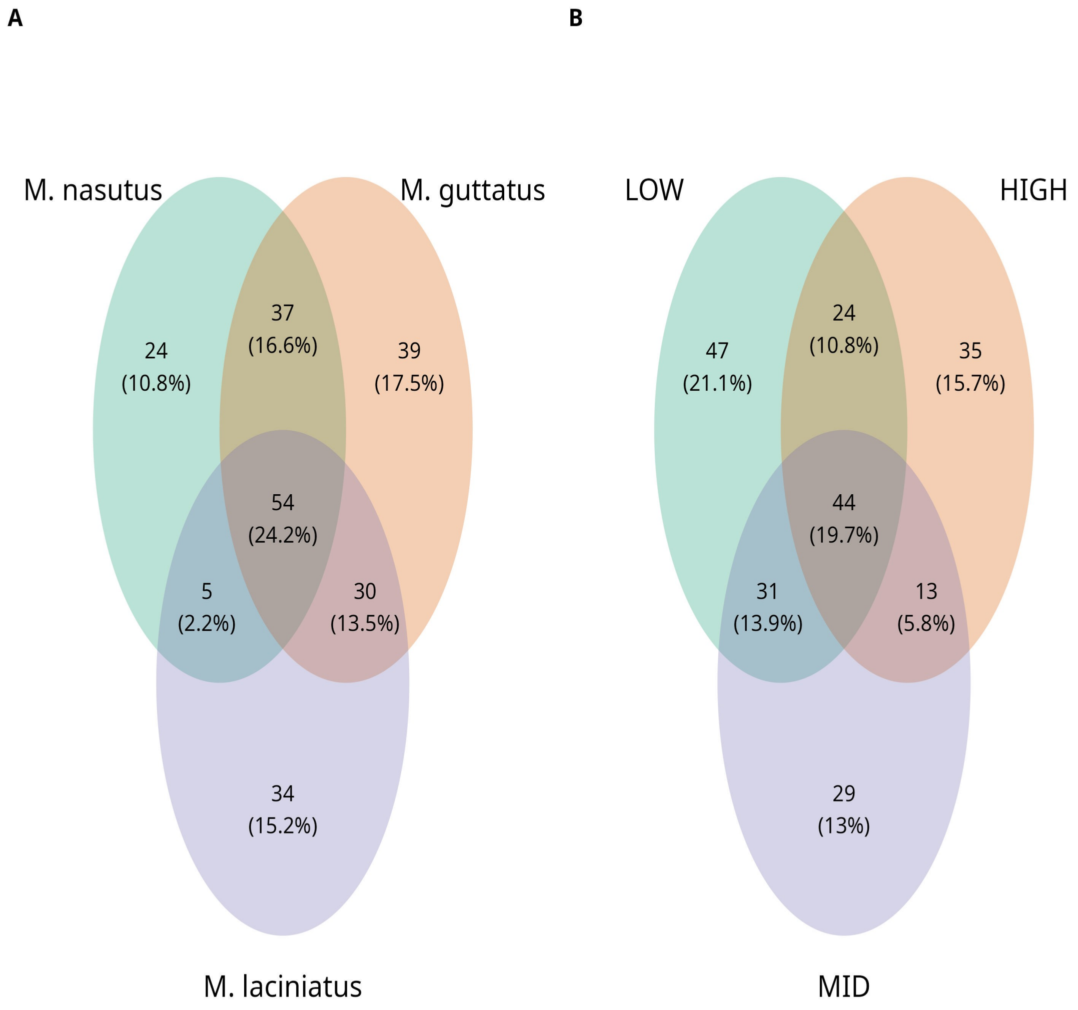

Figure S3. Venn diagrams of presence and absence of ASVs among species and elevation zones un-rarified data and rarified data. A) Represents the overlap of ASVs in in host species and B) per elevation zone. Number of ASVs are represented the count and in parentheses as a ratio shared between host species and elevation zone

13.4 Figure S4

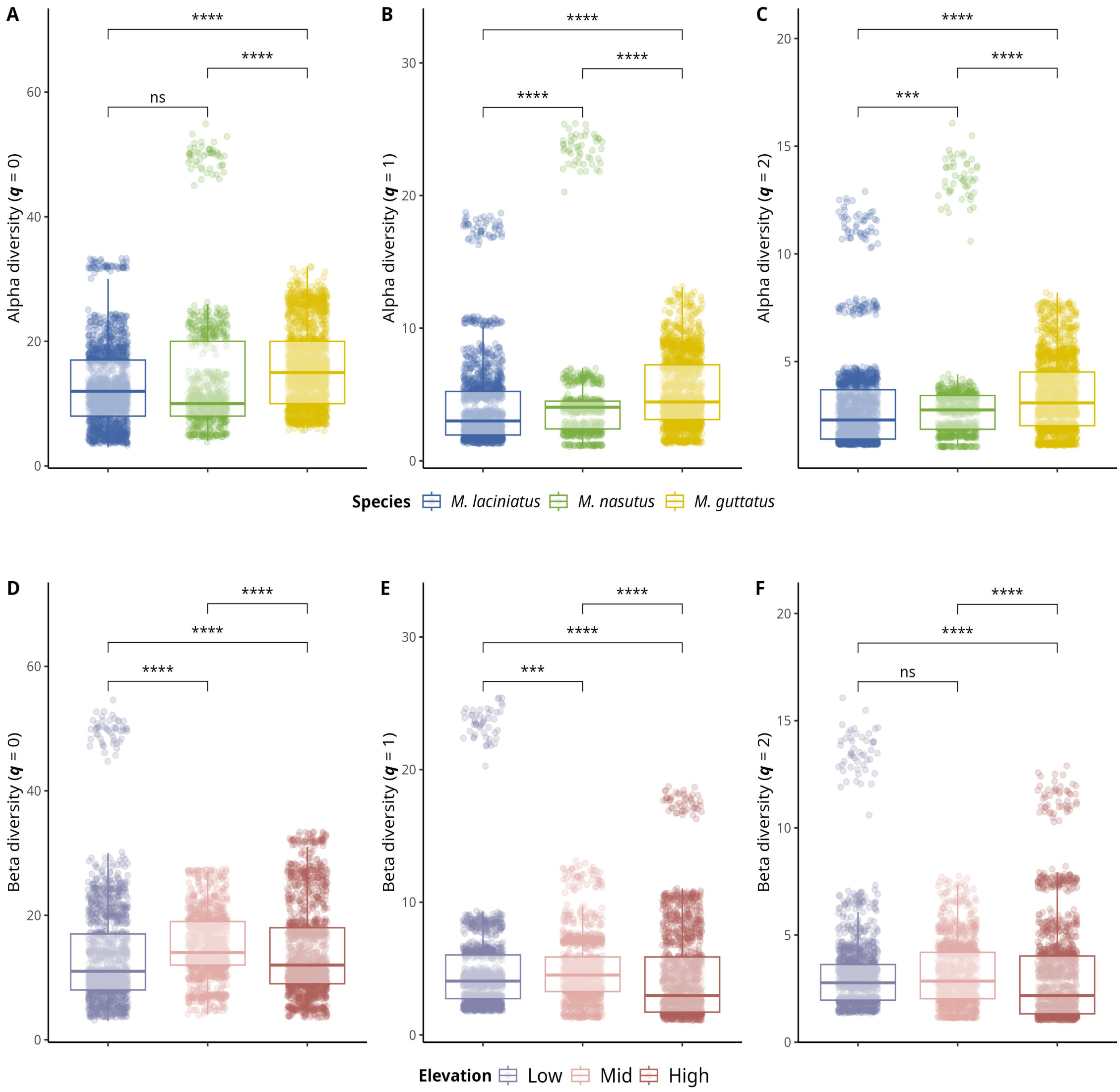

Figure S4. Alpha and beta diversity means comparisons in host species and elevation zones. Alpha diversity means comparisons; A) Observed ASV richness (q = 0); B) Shannon's entropy (q = 1); and C) Inverse Simpson's index (q = 2) per host species. A-C) Blue filled boxplots correspond to *M. laciniatus*, green filled to *M. nasutus*, and yellow to *M. guttatus*. Beta diversity mean comparisons; A) Observed ASV richness (q = 0); B) Shannon's entropy (q = 1); and C) Inverse Simpson's index (q = 2) per elevation zone. D-F) Violet boxplots correspond to LOW elevation sites, pink filled to MID and light maroon to HIGH elevation sites, while squares represent *M. laciniatus*, circles *M. nasutus* and triangles *M. guttatus*. Significance levels are represented by *ns* (not significant) and asterisks [p < .05 (*), p < .01 (**), p < .001 (***), and p < .0001 (****)].

13.5 Figure S5

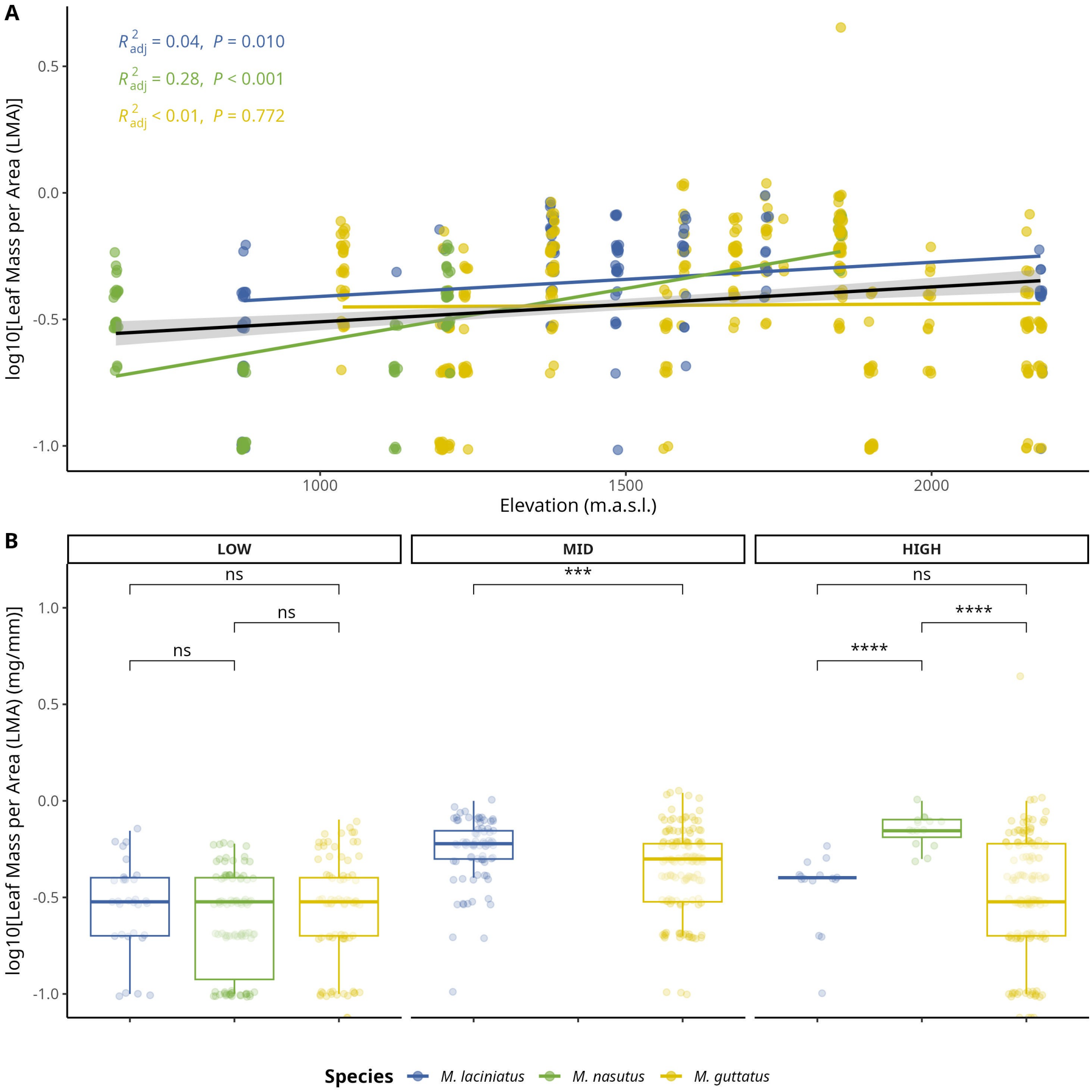

Figure S5. Comparison of log-transformed leaf mass per area (LMA) by species and elevation zone. A) Compares logLMA means by species and elevation zone: LOW (<1220 m), MID (1221 - 1828 m), HIGH (> 1829 m). B) Change in logLMA per species as elevation increases. The black trend line represents the fitted model for all data points, see main text for R^2_{adj} and p values. Significance levels are represented by *ns* (not significant) and asterisks [$p < .05$ (*), $p < 0.01$ (**), $p < .001$ (***), and $p < .0001$ (****)].

13.6 Figure S6

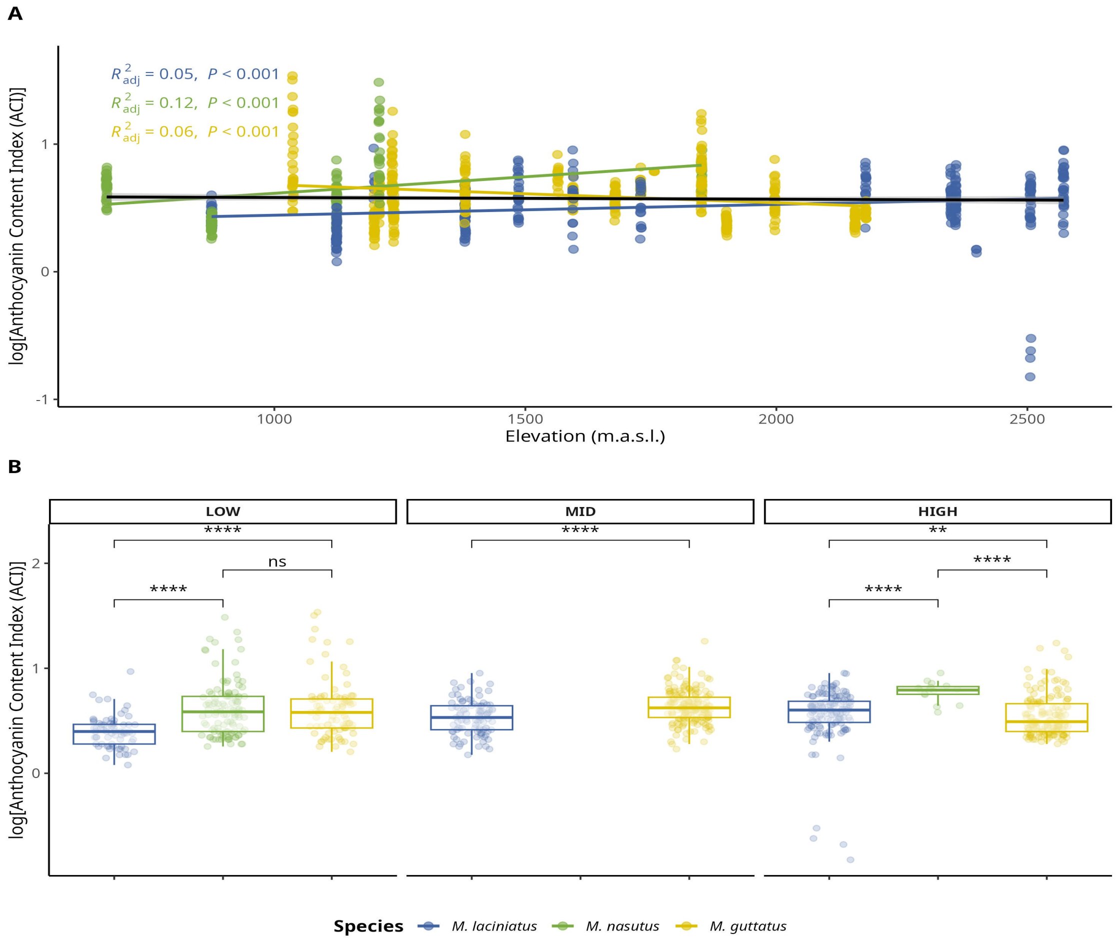

Figure S6. Change in log-transformed anthocyanin content index (ACI) by species and elevation zone. A) Compares logACI means by species and elevation zone: LOW (<1220 m), MID (1221 - 1828 m), HIGH (> 1829 m). The black trend line represents the fitted model for all data points, see main text for R^2_{adj} and *p* values. B) Change in logLPS per species as elevation increases. Significance levels are represented by *ns* (not significant) and asterisks [*p* < .05 (*), *p* < .01 (**), *p* < .001 (***), and *p* < .0001 (****)].

13.7 Figure S7

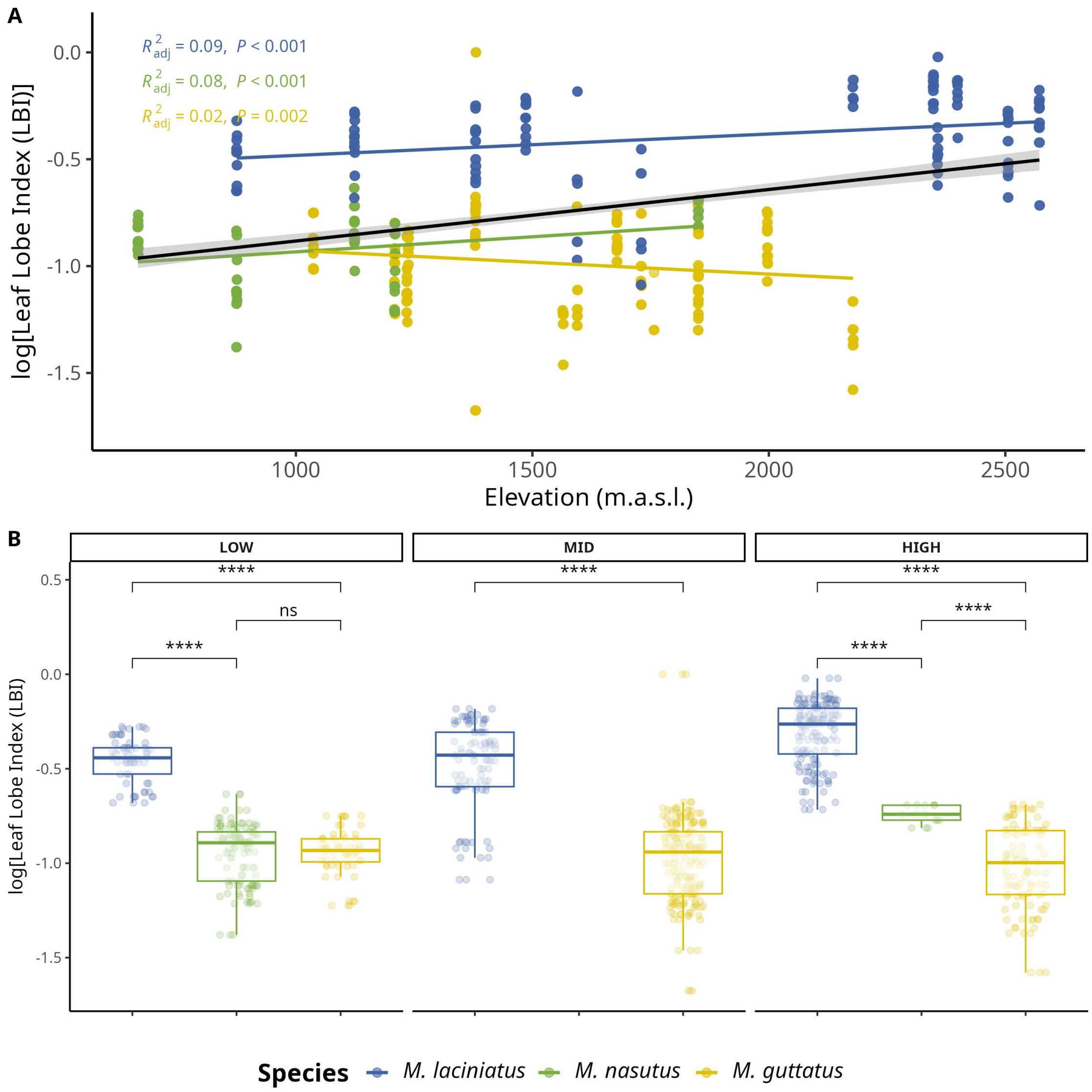

Figure S7. Change in log-transformed leaf lobe index (LBI) by species and elevation zone. A) Compares logLBI means by species and elevation zone: LOW (<1220 m), MID (1221 - 1828 m), HIGH (> 1829 m). The black trend line represents the fitted model for all data points, see main text for R^2_{adj} and *p* values. B) Change in logLPS per species as elevation increases. Significance levels are represented by *ns* (not significant) and asterisks [*p* < .05 (*), *p* < .01 (**), *p* < .001 (***), and *p* < .0001 (****)].

13.8 Figure S8

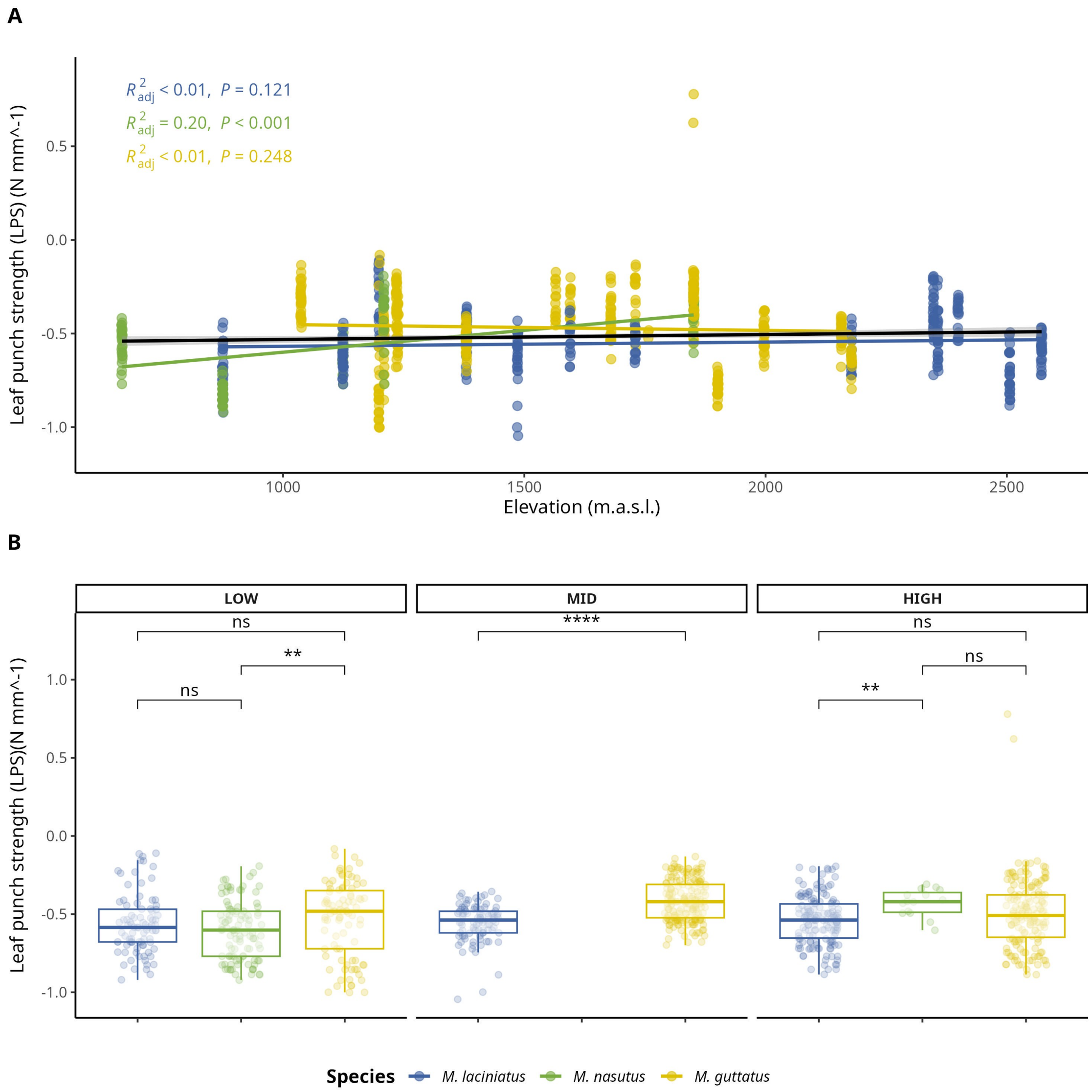

Figure S8. Change in log-transformed leaf-punch strength (LPS), a measure of leaf toughness, by species and elevation zone. A) Compares logLPS means by species and elevation zone: LOW (<1220 m), MID (1221 - 1828 m), HIGH (> 1829 m). The black trend line represents the fitted model for all data points, see main text for R^2_{adj} and p values. B) Change in logLPS per species as elevation increases. Significance levels are represented by *ns* (not significant) and asterisks [p < .05 (*), p < .01 (**), p < .001 (***), and p < .0001 (****)].

13.9 Figure S9

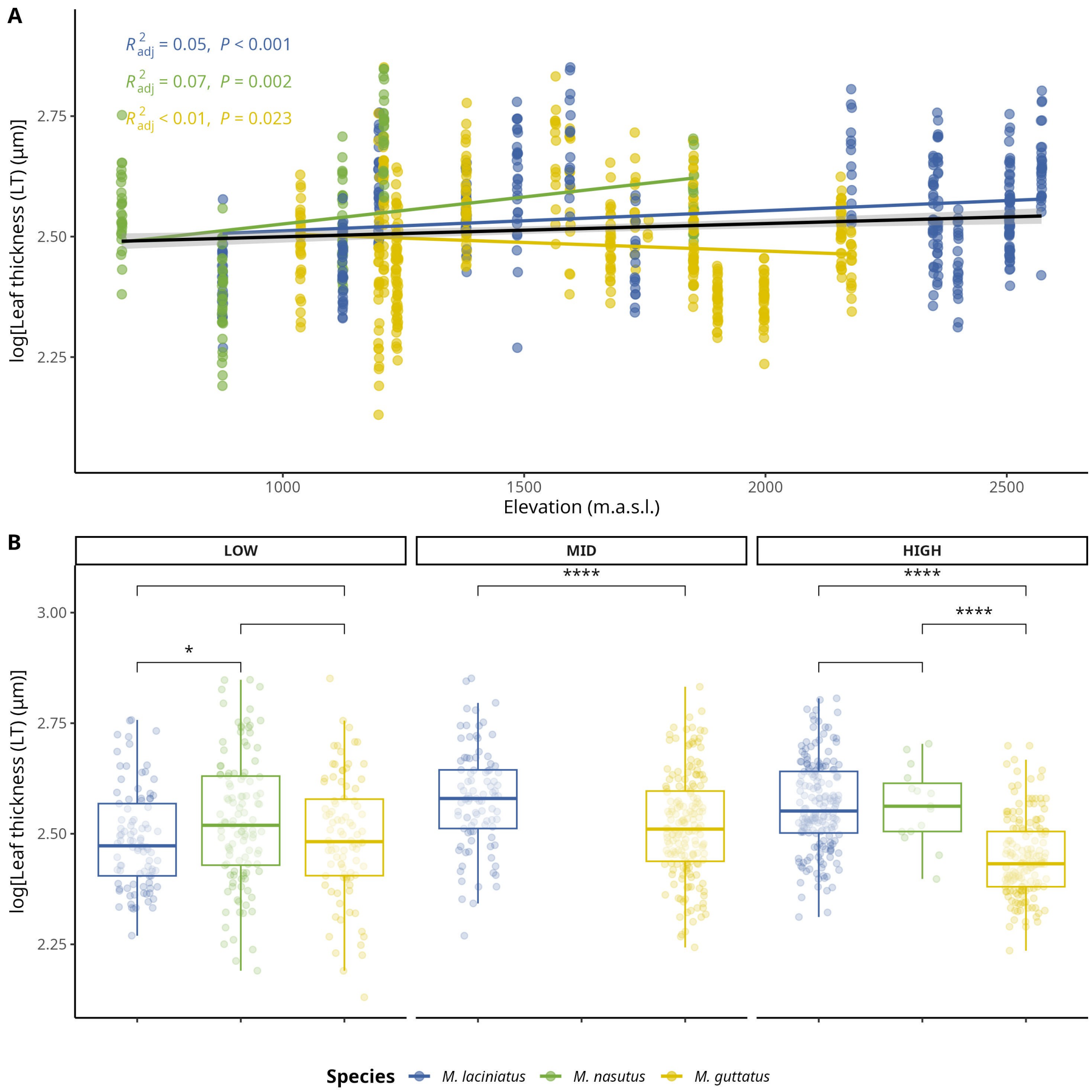

Figure S9. Change in log-transformed leaf thickness (LT) (m) by species and elevation zone. A) Compares logLT means by species and elevation zone: LOW (<1220 m), MID (1221 - 1828 m), HIGH (> 1829 m). The black trend line represents the fitted model for all data points, see main text for R^2_{adj} and p values. B) Change in logLT per species as elevation increases. Significance levels are represented by *ns* (not significant) and asterisks [$p < .05$ (*), $p < .01$ (**), $p < .001$ (***), and $p < .0001$ (****)].

13.10 Table S1

Number of individuals collected per population/site and species.

Site	Site Name	Forest/Park	Latitude	Longitude	*M. laciniatus*	*M. nasutus*	*M. guttatus*
AGFB	Angel Falls	Sierra National Forest	37.338230	-119.570677	11	10	
BHEB	Bald Mountain	Sierra National Forest	37.053544	-119.393233		9	10
BMLZ	Bretz Mill Road	Sierra National Forest	37.090625	-119.314015	5		5
CCNB	Crab Tree Road	Stanislaus National Forest	38.183927	-119.963252			12
CCRB	Carson Creek	Mariposa County	37.486337	-119.997920		9	
CFZZ	Crane Flat	Yosemite National Park	37.755986	-119.802952			10
HDRB	Highland Drive Road	Sierra National Forest	37.288671	-119.534709			10
HHBB	Hetch Hetchy B	Yosemite National Park	37.959056	-119.778553	10		10
HULB	Huntingto n Lake	Sierra National Forest	37.235530	-119.149620	5		5
HUOB	Huntingto n Lake Overview	Sierra National Forest	37.150723	-119.273507		5	17
KNOB	The Knobs	Yosemite National Park	37.853852	-119.440369	10		
LMZZ	Little Meadow	Yosemite National Park	37.767781	-119.772719			10

NFWB	North Fork Willow Creek	Sierra National Forest	37.257397	-119.521415	10	10	
OPNZ	Olmstead Point	Yosemite National Park	37.807232	-119.486313	20		
OSKB	Onion Skins	Yosemite National Park	37.842693	-119.594572	7		
SDSB	Stoddard Spring	Stanislaus National Forest	38.136194	-120.094614			10
SHLZ	Shaver Lake	Sierra National Forest	37.145131	-119.307426	5		5
SPLB	Spotless	Yosemite National Park	37.847194	-119.586129	9		
TRTZ	Turtle Back Dome	Yosemite National Park	37.716098	-119.705317	10		
TWGB	Tawonga	Stanislaus National Forest	37.858820	-119.951305			9
WCFB	Wildcat Falls	Yosemite National Park	37.723593	-119.719077	11		12
WSHB	Wawona School	Yosemite National Park	37.543343	-119.647878			12
YCNB	Yosemite Creek B	Yosemite National Park	37.843904	-119.572754	10		

13.11 Table S2

Prevalence of phyla in samples per species.

		Occurrence
M. laciniatus		
	Ascomycota	57
	Basidiomycota	62
	Mortierellomycota	2
	Olpidiomycota	2
M. nasutus		
	Ascomycota	29
	Basidiomycota	27
	Chytridiomycota	2
	Mortierellomycota	2
	Rozellomycota	2
M. guttatus		
	Ascomycota	58
	Basidiomycota	63
	Mortierellomycota	2
	Olpidiomycota	1

Occurrence represents the number of samples where each phylum was detected per species.

13.12 Table S3

Term	Df	Sum. of Sqs.	F-value	p-value
logLBI	1.000	57.693	306.561	***
Elevation_m	1.000	53.745	285.579	***
Species	2.000	29.173	77.508	***
Elevation_m:Species	2.000	45.614	121.188	***
Residual	3,443.000	647.959		

Significance levels are represented by ns (not significant) and asterisks [p < .05 (*), p < .01 (**), p < .001 (***), and p < .0001 (****)].

13.13 Supplementary methods

13.13.1 Illumina TruSeq adapters

The modified primers for the first PCR (adapter ligation and ITS1 amplification) were as follows:

5’-CAC TCT TTC CCT ACA CGA CGC TCT TCC GAT CTC TTG GTC ATT TAG AGG AAG

TAA-3’ (forward) and 5’-GTG ACT GGA GTT CAG ACG TGT GCT CTT CCG ATC TGC TGC GTT CTT CAT CGA TGC-3’ (reverse). See (Gardes and Bruns, 1993) and (White, T. J. et al., 1990) for more details.

Supplementary Literature Cited

Gardes, M., and T. D. Bruns. 1993. ITS primers with enhanced specificity for basidiomycetes - application to the identification of mycorrhizae and rusts. *Molecular Ecology* 2: 113–118.

White, T. J., Bruns, T.D., S. B. Lee, and J. W. Taylor. 1990. Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics. 315–322.